

IF ELSE AND BOOLEANS:-

Booleans :-

A Boolean is just a value that can be :- true or false

```
let isLoggedIn = true;
let isAdmin = false;
```

If statements in JavaScript:-

syntax:-

```
if(condition){
    //runs if condition is true
}
```

Example:-

```
let age = 17;

if(age >= 18){    // 17 >= 18
    console.log("You are able to vote!")
}
```

2) if .. else

syntax:-

```
if(condition){
    //runs if condition is true
}else{
    // runs if condition is false
}
```

Example:-

```
let age = 17;

if(age >= 18){    // 17 >= 18
    console.log("You are able to vote!")
}else{
    console.log("You are not able to vote")
}
```

3) if .. else if .. else

```
let marks = 89;
let grade = '';

if (marks >= 90) {
  grade = 'A+';
} else if (marks >= 85 && marks < 90) {
  grade = 'A';
} else if (marks >= 80 && marks < 85){
  grade = 'B+';
} else if (marks >= 70 && marks < 80){
  grade = 'B';
} else {
  grade = 'F';
}
console.log(`You got ${grade}`);
```

Output :- You got A

Comparison Operator:-

==	equal
===	strict equal
!=	not equal
>	greater than
<	less than
>=	greater or equal
<=	less or equal

Logical operators:-
&&, ||, ^

Rock paper Scissor project:-

Algorithm:-

- 1) save users move :- rock, paper, scissor
- 2) Computer picks a move randomly :- rock(0.00 --> 0.33), paper(0.33 --> 0.66), scissors(0.66 --> 0.99)
- 3) compare both user and computer moves
- 4) return the result

Truthy and falsy values

All the values which have something stored inside them are truthy. JS by default converts all values to truthy values, except some exceptions which are known as falsy values.

Falsy values example:-
"", 0, -0, undefined, null, NaN

Ternary Operator:-

Ternary operator is a shortcut for If else

Syntax :-
let var1 = condition ? "If condition is true" : "If condition is false" ;

Example 1 :-
const age = 13;
let canJoinOurClub = age >= 18 ? "Yes" : "No";
console.log(canJoinOurClub);
Output :- No

Example 2 :-
const isLoggedIn = true; // if logged in then print "Welcome" else print "Log in first"

let status = isLoggedIn ? "Welcome" : "Login First";
console.log(status);

Output :- "Welcome".

Nested Ternary

Possible but not can messy

Example:-

```
let marks = 78;  
let grade = marks >= 90 ? "A" :  
               marks >= 80 ? "B" :  
               marks >= 70 ? "C" :  
               "F";  
console.log(grade);
```

Output : B

Guard operators :-

Guard operators are used to "Guard" from unwanted values

Example :-

- 1) &&
- 2) ||
- 3) ??

1) AND OPERATOR(&&) :-

Returns first falsy value, if all truthy returns last value.

Example :

```
// console.log("hello" && 123);
```

Output :- 123

Example:-

```
// let apiFetch = "succefully";  
// let data = {  
//   name : "Karan",  
//   age : 23  
// };  
// let dataFetched = apiFetch && data;  
// if(dataFetched){  
//   console.log(dataFetched.age);  
// }else{  
//   console.log("Can't fetch data");  
// }
```

Output :- 23

2) OR Operator(||)
Returns first truthy value

Example :-

```
let userName = undefined || "Guest";  
console.log(userName);
```

Output :- Guest

3) Nullish Coalescing Operator(??)

Returns right side only if left side is null or undefined.

Example :-
let count = 0;
console.log(count ?? 100);

Output :- 0

let count = null;
console.log(count ?? 100);

Output :- 100